Week 1 Homework Solutions

1.1

9 Let us organize the information.

	Acme	Nadir	Quixote
revenue	138	87	111
profit	8	5	13

Immediately we can check that (a) is false, (b) is true, and (c) is true. For (d), recall that "if P then Q" is satisfied if P is false, or if P is true and Q is true. The condition P is not satisfied, so the statement is automatically true ("Quixote Media" had the lowest net profit" is false). For (e), recall that "P if and only if Q" is true when P and Q are both true or both false. Both statements are true, so it evaluates to true.

12bdf (b) The election is decided or the votes have been counted. (d) If the votes have been counted then the election is decided. (f) If the election is not decided then the votes have not been counted.

16bdf (b) $p \wedge q \wedge r$ (d) $p \wedge \neg q \wedge r$ (f) $r \leftrightarrow (q \vee p)$ (since inclusive OR makes more sense in this context).

18ab (a) True, since both sides are T. (b) False, since the left side is T but the right side is F

20ab (a) True, since the hypothesis is F. (b) True, since the hypothesis is F.

30 (a) Converse: If I will stay it home, it will snow tonight. *I added the word will to be consistent with English grammar, and it is implicit that tonight is in the future here.*
Contrapositive: If I won't stay at home, it will not snow tonight.

(b) Converse: If I go to the beach then it is a sunny summer day. *or: It is a sunny summer day whenever I go to the beach.*
Contrapositive: If I don't go to the beach, it is not a sunny summer day.

(c) Converse: If I sleep until noon, then I must have stayed up late.
Contrapositive: If I don't sleep until noon, I must not have stayed up late.

33 See the solutions in the back of the book (page S1).

52 The assertion "this statement is false" does not have a well-defined true/false value (because it is paradoxical), so fails to satisfy the definition of a proposition.

1.2

23 A is a knight and B is a knave.

27 A is a knave and B is a knight.

1.3

12(a) We use the following truth table:

p	q	$p \vee q$	$\neg p \wedge (p \vee q)$	$(\neg p \wedge (p \vee q)) \rightarrow q$
T	T	T	F	T
T	F	T	F	T
F	T	T	T	T
F	F	F	F	T

12(b) We use the following truth table:

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	T	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

12(c) We use the following truth table:

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

12(d) We use the following truth table:

p	q	r	$p \vee q$	$q \rightarrow r$	$p \rightarrow r$	$((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	T	T	T	T
T	F	F	T	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	F	T	T	T
F	F	F	F	T	T	T

14 Show that each conditional statement in Exercise 12 is a tautology using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false.

(a) If the hypothesis is true and conclusion is false, then p and q are both false. But then, $p \vee q$ is false as well, making the hypothesis false.

(b) Suppose the hypothesis is true and the conclusion is false. Then p is true and r is false. But then applying p to the hypothesis gives us that q and then r are true, which is a contradiction.

(c) Suppose the hypothesis is true and the conclusion is false. Then, p and true and q is false. But that just means that $p \to q$ is false, which is a contradiction.

(d) Suppose the hypothesis is true and the conclusion is false. Then in particular r is false, which requires that p and q are also false. But then $p \lor q$ is false, which is a contradiction.

16 Show that each conditional statement in Exercise 12 is a tautology by applying a chain of logical identities.

(a) $(\neg p \land (p \lor q)) \to q \equiv \neg(\neg p \land (p \lor q)) \lor q \equiv p \lor \neg(p \lor q) \lor q \equiv (p \lor q) \lor \neg(p \lor q) \equiv T$

(b) $((p \to q) \land (q \to r)) \to (p \to r) \equiv \neg((p \to q) \land (q \to r)) \lor \neg p \lor r \equiv \neg(p \to q) \lor \neg(q \to r) \lor \neg p \lor r \equiv (p \land \neg q) \lor (q \land \neg r) \lor \neg p \lor r \equiv (\neg p \lor (p \land q)) \lor (q \land \neg r) \lor r \equiv ((\neg p \lor p) \land (\neg p \lor q)) \lor ((q \lor r) \land (\neg r \lor r)) \equiv (T \land (\neg p \lor q)) \lor ((q \lor r) \land T) \equiv (\neg p \lor q) \lor (q \lor r) \equiv q \lor \neg q \lor \neg p \lor r \equiv T \lor \neg p \lor r \equiv T$

(c) $(p \land (p \to q)) \to q \equiv \neg(p \land (p \to q)) \lor q \equiv \neg p \lor \neg(p \to q) \lor q \equiv (\neg p \lor (p \land \neg q)) \lor q \equiv ((\neg p \lor p) \land (\neg p \lor \neg q)) \lor q \equiv (T \land (\neg p \lor \neg q)) \lor q \equiv \neg p \lor \neg q \lor q \equiv T$

(d) $((p \lor q) \land (p \to r) \land (q \to r)) \to r$
$\equiv \neg((p \lor q) \land (p \to r) \land (q \to r)) \lor r$
$\equiv \neg(p \lor q) \lor \neg(p \to r) \lor \neg(q \to r) \lor r$
$\equiv (\neg p \land \neg q) \lor (p \land \neg r) \lor (q \land \neg r) \lor r$
$\equiv (\neg p \land \neg q) \lor (p \land \neg r) \lor ((q \lor r) \land (\neg r \lor r))$
$\equiv (\neg p \land \neg q) \lor (p \land \neg r) \lor ((q \lor r) \land T)$
$\equiv (\neg p \land \neg q) \lor (p \land \neg r) \lor q \lor r$
$\equiv (\neg p \land \neg q) \lor q \lor (p \land \neg r) \lor r$
$\equiv ((\neg p \lor q) \land (\neg q \lor q)) \lor ((p \lor r) \land (\neg r \lor r))$
$\equiv (\neg p \lor q) \lor (p \lor r) \equiv \neg p \lor p \lor q \lor r \equiv T$

1.4

10ace Let $C(x)$ be the statement "x has a cat,", let $D(x)$ be the statement "x has a dog,", and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x), D(x), F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

(a) $\exists x\,(C(x) \land D(x) \land F(x))$

(c) $\exists x\,(C(x) \land F(x) \land \neg D(x))$

(e) $\exists x \exists y \exists z\,(C(x) \land D(y) \land F(z))$

16 Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

(a) True, becuase the positive or negative square root of 2 satisfies this.
(b) False, since the square of any real number is nonnegative.
(c) True. The statement is equivalent to $\forall x(x^2 \geq -1)$.
(d) False; consider $x = 0$ or 1.

36ab Express the negation of each of these statements in terms of quantifiers without using the negation symbol.

(a) $\exists x\,(x \leq -2 \vee x \geq 3)$
(b) $\exists x\,(x < 0 \vee x \geq 5)$

38 Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers.

(a) $x = 0$ or 1
(b) $x = \sqrt{2}$
(c) $x = 0$

46 They are not logically equivalent. For example, if the domain is the set of real numbers and $P(x)$ says $x < 2$ and $Q(x)$ says $x^2 \geq 4$, we get that $\forall x\, P(x)$ and $\forall x\, Q(x)$ are both false, and so $\forall x\, P(x) \leftrightarrow \forall x\, Q(x)$ is true. But obviously the statement $\forall x\,(P(x) \leftrightarrow Q(x))$ is false.

52 The same counterexample above applies here. Note that $\forall x\,(P(x) \vee Q(x))$ is true, but $\forall x\, P(x) \vee \forall x\, Q(x)$ is false.

1.5

20 Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.

(a) $\forall x \forall y\,((x < 0 \wedge y < 0) \rightarrow x \cdot y > 0)$
(b) $\forall x \forall y\,((x > 0 \wedge y > 0) \rightarrow \frac{x+y}{2} > 0)$
(c) $\neg \forall x \forall y\,((x < 0 \wedge y < 0) \rightarrow x - y < 0)$
(d) $\forall x \forall y\,(|x + y| \leq |x| + |y|)$

24abd Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

(a) There is a real number x that, when added to any other real number y, returns the same value y.
(b) Subtracting a negative number from a nonnegative number yields a positive number.

(d) The product of nonzero numbers is nonzero.

44 We translate this into the statement that if x, y, z are roots of a quadratic polynomial, some pair of them must be equal.

$\forall a, b, c\, \forall x, y, z\, ((a \neq 0 \wedge ax^2+bx+c = 0 \wedge ay^2+by+c = 0 \wedge az^2+bz+c = 0)$
$\rightarrow (x = y \vee x = z \vee y = z))$

1.6

16 For each of these arguments determine whether the argument is correct or incorrect and explain why.

(a) The argument has the following structure:

i. $\forall x\, (U(x) \rightarrow D(x))$
ii. $\neg D(m)$
iii. $\therefore \neg U(m)$

It is valid because we can instantiate the first premise to get $U(m) \rightarrow D(m)$ and then apply Modus Tollens.

(b) The argument has the following structure:

i. $c \rightarrow \neg d$
ii. $\neg c$
iii. $\therefore \neg d$

It's not hard to check on a truth table that it is invalid.

(c) This argument is similarly invalid.

(d) This argument is valid. The structure is like the first one but with Modus Ponens instead of Modus Tollens.

20 (a) This argument is invalid because we can choose a to be negative, making the premise true and conclusion false.

(b) This argument is valid.

24 Steps 3 and 5, labeled "simplification" are invalid. The argument tries to conclude $P(c)$ from $P(c) \vee Q(c)$, but it is possible that $P(c)$ is false and $Q(c)$ is true, in which case $P(c) \vee Q(c)$ is true.